

Signal and System Design (IGS2137)

Jump Together, Fly Farther!



인하대학교 국제학부

Week 2 Lab

**ISE Department
Prof. Mehdi Pirahandeh**

- Introduction to MATLAB
- Creating Variables
- Some Useful MATLAB Functions
- Data Types
- Q & A
- Survey

What is MATLAB®?

MATLAB® /Simulink® is a powerful software tool for:

- Performing mathematical computations and signal processing
- Analyzing and visualizing data (excellent graphics tools)
- Modeling physical systems and phenomena
- Testing engineering designs

- **Aircraft/Defense:** control and guidance system design and simulation, communications
- **Robotics:** design and control
- **Automotive:** cruise control, stability enhancement, fuel injection systems, hybrid power-train, sound suppression ...
- **Communications:** voice over internet, cell-phone, satellite, antenna design, wireless, error coding ...
- **Biotech, Pharmaceutical, Medical:** drug discovery and development, imaging procedures, cancer diagnosis ...

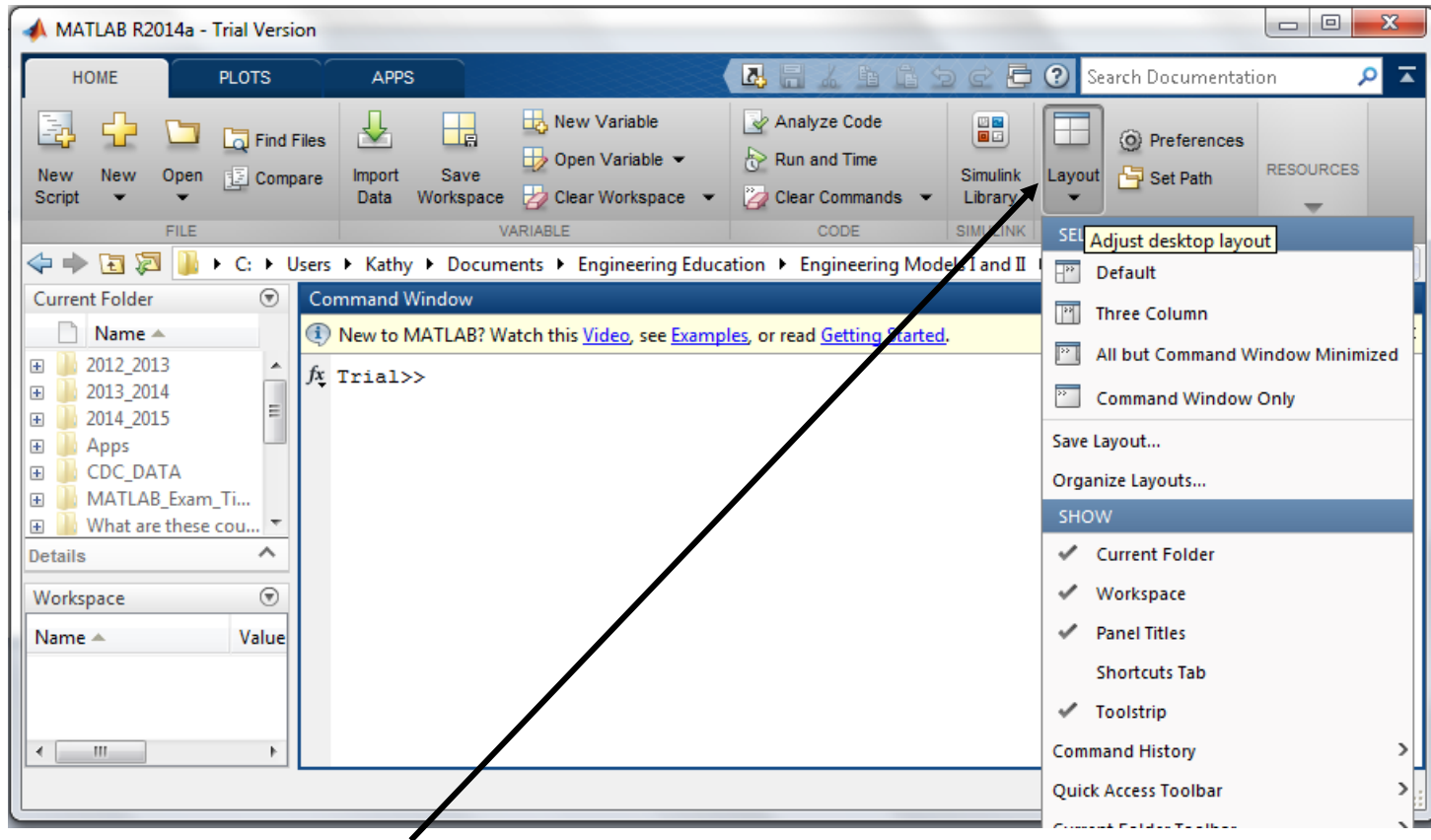
Industry Applications (con't)

- **Electronics:** chip design, acoustics, voice processing and recognition
- **Industrial Automation and Machinery:** sensor design, machinery design and control
- **Utilities and Energy:** power conversion and control
- **Computers:** security systems, printer design
- **Financial:** portfolio management and risk, commodity trading, currency markets

MATLAB® DESKTOP

- The **Command window** is where you type MATLAB commands following the prompt: `EDU>>`
- The **Workspace window** shows all the variables you have defined in your current session. Variables can actually be manipulated within the workspace window.
- The **Command History** window displays all the MATLAB commands you have used recently – even includes some past sessions.
- The **Current Folder** window displays all the files in whatever folder you select to be current.

MATLAB Desktop



- You can select what is on your desktop by Clicking on **Layout**.
- Go down to Command History and select docked.

USING MATLAB® AS A CALCULATOR

Arithmetic Operators and Order of Operations

- Addition (+), Subtraction (-), Multiplication (*), Division (/), Power (^)
- Order of Operations (same rules you should already know from math class and using a calculator)
 1. Complete all calculations inside parenthesis or brackets using the precedent rules below
 2. Powers (left to right)
 3. Multiplication and Division (left to right)
 4. Addition and Subtraction (left to right)

Arithmetic Operators and Order of Operations

Some Examples:

EDU>> $10/5*2$

EDU>> $5*2^3+4(2)$

EDU>> -1^4

EDU>> $8^{1/3}$

Exercise 1

Calculate the area and circumference of a circle with a radius of 4 cm.

$$\text{Area} = 50.27 \text{ cm}^2$$

$$\text{Circumference} = 25.13 \text{ cm}$$

CREATING AND USING VARIABLES

- In the previous exercise, you probably typed in a value for pi.
- MATLAB has a built-in variable for pi. Try this:

```
EDU>> pi
```

Variables

You can even create your own variables. Try this:

```
EDU>> radius = 4
```

What do you see in your workspace window?

Now try this:

```
EDU>> area = pi*radius^2
```

What do you see in your workspace window now?

Naming Rules for Variables

1. Variable names must begin with a letter
2. Names can include any combinations of letters, numbers, and underscores
3. Maximum length for a variable name is 63 characters
4. MATLAB® is case sensitive. The variable name A is different than the variable name a.
5. Avoid the following names: i, j, pi, and all built-in MATLAB® function names such as length, char, size, plot, break, cos, log, ...
- 6.) It is good programming practice to name your variables to reflect their function in a program rather than using generic x, y, z variables.

Creating Variables & Assigning Values



인하대학교
INHA UNIVERSITY

At the MATLAB command prompt (EDU>>) type:

x = 10.57;

What happens? (Hint: look at workspace window)

Several things happen with this simple MATLAB command:

- A variable, **x**, of type double is created
- A memory location for the variable **x** is assigned
- The value **10.57** is stored in that memory location called **x**.

Creating Variables & Assigning Values



인하대학교
INHA UNIVERSITY

At the MATLAB command prompt (EDU>>) type:

$$x = 73.65$$

What happens?

- The old value for **x** (10.57) is replaced by the new value (**73.65**)
- Also, since the semicolon was left off the end, we see the result in the command window (as well as in the workspace window)

Exercise 2

1. In MATLAB create two variables: $a = 4$ and $b = 17.2$
2. Now use MATLAB to perform the following set of calculations:

$$5a$$

$$\sqrt[3]{b + 9.8}$$

$$10\sqrt{5a + 16}$$

Creating Strings (Text Variables)

Variables do not have to be *numbers*. At the MATLAB command prompt type:

```
month = 'August'
```

What happens?

Several things happen with this simple MATLAB command:

- A variable, **month**, of type string (character array) is created
- A memory location for the variable **month** is assigned
- The string **August** is stored in that memory location called **month**.
- Notice that to enter a string, we must put single quotes around it

Displaying Variables

We can display a variable (i.e., show its value) by simply typing the name of the variable at the command prompt (leaving off the semicolon). Try this!

We can also use a function called **disp** to display variables. Type the following commands at the command prompt:

```
>> disp('The value of x is:'); disp(x)
```

SOME USEFUL MATLAB FUNCTIONS

Some MATLAB® Math Functions

Function	MATLAB®		Function	MATLAB®
<i>cosine</i>	cos or cosd		<i>square root</i>	sqrt
<i>sine</i>	sin or sind		<i>exponential</i>	exp
<i>tangent</i>	tan or tand		<i>logarithm (base 10)</i>	log10
<i>cotangent</i>	cot or cotd		<i>natural log (base e)</i>	log
<i>arc cosine</i>	acos or acosd		<i>round to nearest integer</i>	round
<i>arc sine</i>	asin or asind		<i>round down to integer</i>	floor
<i>arc tangent</i>	atan or atand		<i>round up to integer</i>	ceil
<i>arc cotangent</i>	acot or acotd			

Note: $\cos(\alpha)$ assumes α in radians; whereas, $\cosd(\alpha)$ assumes α in degrees.
 $\text{acos}(x)$ returns the angle in radians; whereas, $\text{acosd}(x)$ returns the angle in degrees.

$$\pi \text{ radians} = 180 \text{ degrees}$$

Other Useful Stuff

- **clear** clears all variables in the MATLAB® workspace
- **clear a, b** just clears variables a and b
- **clc** clears the command window
- **i** and **j** are defined in MATLAB to be $\sqrt{-1}$.
- If you define these variables to be something else, you lose the ability to work with complex numbers. So, avoid using i and j as variable names.
- **pi** is defined in MATLAB as **3.14159....**

Help!

- The **help** command provides information about a function. Type **help cos** at the command prompt. This only works if you know the name of the function you want help with.
- You can also click on Help in the MATLAB toolbar and search for information by keyword(s).

HOW COMPUTERS STORE VARIABLES

How Computers Store Variables

Suppose we type the following commands in MATLAB:

```
EDU>> y = 42;
```

```
EDU>> FavoriteDay = 'Friday'
```

We know MATLAB stores the values associated with the variables, `y` and `FavoriteDay`, in memory.

How are these values stored?

How Computers Store Variables

Computers store all data (numbers, letters, instructions, ...) as strings of 1s and 0s (bits).

A **bit** is short for **binary digit**. It has only two possible values: On (1) or Off (0).

Terminology

A **bit** is short for **b**inary digit. It has only two possible values: On (1) or Off (0).

A **byte** is simply a string of 8 bits.

A **kilobyte** (kB) is 1000 bytes

A **megabyte** (MB) is 1,000,000 bytes

A **gigabyte** (GB) is 1,000,000,000 bytes

For a sense of size, click on link below:

<http://highscalability.com/blog/2012/9/11/how-big-is-a-petabyte-exabyte-zettabyte-or-a-yottabyte.html>

Numeric Data Types

MATLAB has several different options for storing numbers as bits. Unless you specify otherwise, all numbers in MATLAB are stored as doubles.

Name	Description	Range
<i>double</i>	64 bit floating point	-1.79769313486232E308 to -4.94065645841247E-324 4.94065645841247E-324 to 1.79769313486232E308
<i>single</i>	32 bit floating point	-3.402823E38 to -1.401298E-45 1.401298E-45 to 3.402823E38
<i>uint8</i>	8 bit unsigned integer	Integers from 0 to 255
<i>int8</i>	8 bit signed integer	Integers from -128 to 127
<i>uint16</i>	16 bit unsigned integer	Integers from 0 to 65535
<i>int16</i>	16 bit signed integer	Integers from -32768 to 32767
<i>uint32</i>	32 bit unsigned integer	Integers from 0 to 4294967295
<i>int32</i>	32 bit signed integer	Integers from -2147483648 to 2147483647

Data Types:

int8, uint8, int16, uint16, int32, uint32

- These data types work for integers as long as the integers don't exceed the range for the data type chosen.
- They take up less memory space than doubles.
- They don't work for non-integers. If you create a variable that is an int8 and try to assign it a value of 14.8, that variable will be assigned a value of 15 instead (closest integer within the range).
- One common application for integer data types is image data (jpeg, png, ...)

Data Types: double and single

- A double uses 64 bits to store a number.
- A single uses 32 bits to store a number.
- Doubles and singles can be used to represent both integers and non-integers.

Why should I care how data is stored in a computer?

Example: Perform each of the following calculations in your head.

$$a = 4/3$$

$$b = a - 1$$

$$c = 3*b$$

$$e = 1 - c$$

What does MATLAB get?

Why should I care how data is stored in a computer?

What does MATLAB get?

$$a = 4/3 = 1.3333$$

$$b = a - 1 = 0.3333$$

$$c = 3*b = 1.0000$$

$$e = 1 - c = 2.2204e-016$$



It is not possible to perfectly represent all real numbers using a finite string of 1s and 0s.

- Not all numbers can be represented exactly even using 64 bit doubles. If we do many, many calculations with numbers which are just a tiny bit off, that error can grow very, very large depending on the type of computations being performed.
- 64 bit doubles have a huge, but still limited range. What happens if we exceed it? Try the following:

```
EDU>> a = 373^1500
```

```
EDU>> b = factorial(172)
```

ASCII Code

When you press a key on your computer keyboard, the key that you press is translated to a binary code.

A = 1000001 (Decimal = 65)

a = 1100001 (Decimal = 97)

0 = 0110000 (Decimal = 48)

ASCII Code

Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
0	00	Null	32	20	Space	64	40	@	96	60	`
1	01	Start of heading	33	21	!	65	41	A	97	61	a
2	02	Start of text	34	22	"	66	42	B	98	62	b
3	03	End of text	35	23	#	67	43	C	99	63	c
4	04	End of transmit	36	24	\$	68	44	D	100	64	d
5	05	Enquiry	37	25	%	69	45	E	101	65	e
6	06	Acknowledge	38	26	&	70	46	F	102	66	f
7	07	Audible bell	39	27	'	71	47	G	103	67	g
8	08	Backspace	40	28	(	72	48	H	104	68	h
9	09	Horizontal tab	41	29	)	73	49	I	105	69	i
10	0A	Line feed	42	2A	*	74	4A	J	106	6A	j
11	0B	Vertical tab	43	2B	+	75	4B	K	107	6B	k
12	0C	Form feed	44	2C	,	76	4C	L	108	6C	l
13	0D	Carriage return	45	2D	-	77	4D	M	109	6D	m
14	0E	Shift out	46	2E	.	78	4E	N	110	6E	n
15	0F	Shift in	47	2F	/	79	4F	O	111	6F	o
16	10	Data link escape	48	30	0	80	50	P	112	70	p
17	11	Device control 1	49	31	1	81	51	Q	113	71	q
18	12	Device control 2	50	32	2	82	52	R	114	72	r
19	13	Device control 3	51	33	3	83	53	S	115	73	s
20	14	Device control 4	52	34	4	84	54	T	116	74	t
21	15	Neg. acknowledge	53	35	5	85	55	U	117	75	u
22	16	Synchronous idle	54	36	6	86	56	V	118	76	v
23	17	End trans. block	55	37	7	87	57	W	119	77	w
24	18	Cancel	56	38	8	88	58	X	120	78	x
25	19	End of medium	57	39	9	89	59	Y	121	79	y
26	1A	Substitution	58	3A	:	90	5A	Z	122	7A	z
27	1B	Escape	59	3B	;	91	5B	[	123	7B	{
28	1C	File separator	60	3C	<	92	5C	\	124	7C	
29	1D	Group separator	61	3D	=	93	5D	]	125	7D	}
30	1E	Record separator	62	3E	>	94	5E	^	126	7E	~
31	1F	Unit separator	63	3F	?	95	5F	_	127	7F	□

Strings in MATLAB

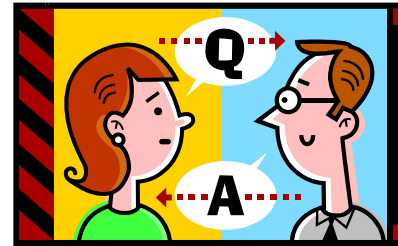
- MATLAB stores strings as an array of characters using the ASCII code.
- Each letter in a string takes up two bytes (16 bits) and the two bytes are the binary representation of the decimal number listed in the ASCII table.

Try the following in MATLAB:

```
EDU>> month = 'August'  
EDU>> double(month)
```

Brief break (if on schedule)

Q&A?



Prof. Mehdi Pirahandeh
E-mail : mehdi@inha.ac.kr